

A local dynamic feature selection fusion method for voice diagnosis of Parkinson's disease

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ABSTRACT

Voice disorders are one of the incipient symptoms of Parkinson's disease (PD). The existing PD voice feature analysis methods generally reduce dimensionality by selecting features with high relevance and low redundancy, ignoring the complementarity between features. To fully explore the intrinsic connection between PD features, this paper proposes a local dynamic feature selection fusion method (LDFSFS) for PD identification and symptom severity prediction. First, the maximal information coefficient is used to calculate the relevance between features and the target value, and removes low relevant features. Then, a dynamic feature selection strategy based on self-organizing map network clustering is designed. This strategy dynamically updates the feature subset in a forward manner, and combines the relevance, redundancy and complementarity for feature selection. Among them, self-organizing map network clustering reduces the computational cost by reducing the number of candidate features, and effectively improves the accuracy of the model. Finally, the selected features are fed into the appropriate classifier or regressor to achieve accurate PD identification and severity progress prediction. Experiments on public OPDD and PTDS datasets show that the classification accuracy, sensitivity and specificity of LDFSFS method on OPDD are 98.20%, 92.00% and 99.50% respectively, and the mean absolute error of predicting PD severity (motor- and total-UPDRS) on PTDS are 1.62 and 1.99 respectively, which are better than the compared methods for the selection of five and eight features respectively, thus indicating that the proposed LDFSFS method can be effectively used in PD voice diagnosis.

1. Introduction

Parkinson's disease (PD) is one of the common neurodegenerative diseases (Ji et al., 2018). The main clinical manifestations of the disease are not only motor characteristics such as trembling, slow movement, postural balance disorder, but also symptoms such as voice disorder and dysphonia (Zhang et al., 2021a). Classification and severity assessment of PD based on voice data has been proven to be an effective approach in recent years (Klumpp et al., 2022; Khan et al., 2020). However, high-dimensional voice features may increase the time complexity of the prediction model, which may affect the accuracy and operational efficiency of the model. Therefore, the selection of a suitable subset of features by mining the relationship between features and the target value and between features is of great significance to improve the accuracy of the prediction model and the early diagnosis and treatment of PD.

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Scholars have put forward a variety of PD voice feature selection (FS) methods. In the PD classification task, Cigdem and Demirel (2018) used the ReliefF method based on Fisher criterion to determine the weights of features, and selected the features with higher weights to input into support vector machine (SVM) for PD classification. Since the ReliefF method ignores the redundancy between features, its selection of highly correlated features will exhibit similar discriminative power, which may degrade the performance of the method (Zhang et al., 2019b; Urbanowicz et al., 2018). Sakar et al. (2019) used the method of minimum redundancy maximum relevance (mRMR) to reduce the redundancy among features to maximize their joint dependency on the target values, so that they constituted a subset of features with maximum joint dependence with the target values, and obtained a more robust and accurate PD classification model. However, the mRMR method is not suitable for processing high-dimensional data due to its high computational cost (Ahmed et al., 2020). Senturk (2020) experimented with two FS methods, feature importance and recursive feature elimination (RFE), respectively, and found that RFE-based SVM outperforms other algorithms and this method can be implemented with less computational cost. Besides considering the relevance and redundancy, Liu et al. (2021) proposed an improved manifold learning algorithm based on the noise and aliasing characteristics of PD voice features, which achieved good performance in PD diagnosis. In the PD severity progression prediction task, various FS methods have also been proposed to select significant features. Tunc et al. (2020) used the Boruta algorithm to determine the significance of features by finding relationships among all relevant features and analyzed the selected features to conclude that vocal fold vibration patterns are important indicators for assessing the severity of PD. Tsanas et al. (2021) evaluated the features using four FS algorithms to obtain features with high importance and tested support vector regression (SVR), extreme gradient boosting (XGBoost) and random forest (RF) regressors respectively. The results show that the prediction performance of RF is always significantly better than that of SVR and XGBoost, which verifies the superiority of RF in PD severity prediction. Despotovic et al. (2020) ranked PD voice feature weights by automatic relevance determination (ARD) method, and selected eight features with the high relevance with the target value. Gaussian process can extend multivariate Gaussian distribution to infinite many random variables, which has been successfully used in latent variable modeling of high-dimensional data. Therefore, the method of combining ARD with Gaussian process can not only reduce the number of PD voice features, but also improve the prediction accuracy of the model.

The above FS methods based on relevance and redundancy have analyzed the intrinsic relationships between PD voice features and combined with suitable classifiers and regressors to obtain good prediction models. However, the FS analysis using only these two properties is not sufficient. This is because a single feature with low classification ability may have a high dependency on other features, that is, there is complementarity between features, and adding them to the feature subset may help to enhance the recognition ability of the model (Chen et al., 2021). Therefore, in order to select effective features and improve the accuracy of the prediction model, it is necessary to analyze the relevance, redundancy and complementarity of features at the same time.

Since redundancy and complementarity can be achieved through the calculation of relevance between features, the selection of an appropriate relevance evaluation method has a significant impact on feature selection. Currently, the Spearman correlation coefficient is often used to evaluate the relevance between PD features (Tsanas et al., 2021). However, its credibility can be significantly reduced when analyzing complex relevance of variables (Zhang et al., 2021c). The maximal information coefficient (MIC) is an effective method to measure the relevance between variables (Reshef et al., 2011). It measures not only linear and nonlinear relationships between variables, but also functional and nonfunctional relationships. In addition, MIC can still give similar scores when there are different types but the same degree of noise in the variables. Therefore, MIC with universality and robustness is more suitable for analyzing the complex relevance between PD voice features.

In order to select important features in combination with the relevance, redundancy and complementarity, this paper designs a dynamic feature selection (DFS) strategy based on clustering, which can dynamically update the importance between candidate features and the selected feature subset. In this paper, the importance of features is calculated by combining the relevance between features and the target value and the redundancy between features. In addition, the strategy uses a specific classifier or regressor to track the prediction results of new feature subsets that are constantly generated in order to effectively eliminate some features with low classification ability and describe the complex dynamic relationships between features in the process of FS. Literature (Despotovic et al., 2020; Tsanas et al., 2021) has shown that Gaussian process and RF exhibit outstanding performance in PD classification and regression tasks, so the DFS strategy uses them as the specific classifier and regressor, respectively. When dealing with high-dimensional data, the above DFS strategy without clustering needs a large computational cost. Clustering the feature set can greatly reduce the number of candidate features in the FS process, which may reduce the complexity of dynamic feature selection method.

Neural networks are widely used because of their powerful nonlinear mapping capabilities. However, neural network-based clustering methods are often limited by unsupervised learning or relying on a large number of labeled samples (Shukla et al., 2020). And the neural network usually requires a large dataset for training, otherwise it is prone to overfitting (Xue et al., 2022). Self-organizing map (SOM) does not have these limitations. SOM is a robust unsupervised clustering algorithm based on competitive learning, and it has been shown to be effective in improving the predictive performance of models in PD voice diagnosis (Nilashi et al., 2018, 2019, 2020). Specifically, the SOM method can automatically divide the feature space into different clusters through nonlinear mapping (Nilashi et al., 2019). The features within the clusters have strong similarity and the number of features in each cluster is relatively small. Therefore, the DFS strategy based on SOM clustering can select the features in each cluster, namely the local feature set, so as to reduce the computational complexity and remove redundant features.

In this paper, a local dynamic feature selection fusion (LDFSf) method is proposed, which performs feature selection based on relevance, redundancy and complementarity, and is applied to PD identification and symptom severity prediction tasks. To verify the effectiveness of the proposed LDFSf model, we conducted experiments on international public OPDD and PTDS datasets, respectively, and compared the prediction results of the proposed method with traditional FS methods as well as the latest available literature. The

experimental results show that the proposed LDFS method can select the feature subset with small number and effective features, and achieve better prediction performance by combining with the appropriate classifier and regressor. Although many feature selection methods based on relevance and redundancy have been studied in [Sakar et al. \(2019\)](#), [Senturk \(2020\)](#), [Tsanas et al. \(2021\)](#), and [Despotovic et al. \(2020\)](#), to the authors' best knowledge, the PD classification and symptom severity prediction methods combining relevance, redundancy and complementarity were not quite considered in the literature. The major contributions of this work are three-fold:

- (1) This paper proposes a novel feature selection method LDFS for PD classification and symptom severity prediction by combining relevance, redundancy and complementarity.
- (2) After selecting strongly relevant features by the MIC, this paper proposes a dynamic feature selection strategy based on SOM clustering to analyze redundancy and complementarity of features. This strategy can further explore the internal relationship among the features, select a small number of features and improve the prediction performance.
- (3) The feasibility and effectiveness of the proposed LDFS method are verified by experiments on the public Oxford Parkinson's disease classification dataset and Parkinson's telemonitoring dataset.

2. Methods

In this paper, the proposed LDFS method consists of three main parts: MIC relevance analysis, DFS strategy based on SOM clustering and classification or regression model prediction. The specific process is shown in [Fig. 1](#). To verify the effectiveness of the proposed LDFS method, experiments were conducted on OPDD classification and PTDS regression datasets respectively. Firstly, the MIC is used to calculate the relevance between PD voice features and the target value because of its ability to find multiple relationships between variables. Next, a dynamic feature selection strategy based on SOM clustering is designed. Among them, the redundancy and complementarity between candidate features and feature subsets are fully considered in DFS, and SOM clustering can reduce the number of candidate features, thus reducing the complexity of DFS. After the selected feature subset is determined, classification and regression models are used to identify PD and predict the symptom severity of PD. For the classification task, the predicted target value is status, where status = 0 represents healthy control (HC) and status = 1 represents the patient with Parkinson's disease (PWP). For the regression task, the target value is the unified Parkinson's disease rating scale (UPDRS), which is widely used internationally to measure the severity of PD symptoms. Specifically, UPDRS is divided into motor-UPDRS and total-UPDRS to describe the motor symptoms and overall symptoms of PD patients, respectively.

2.1. Relevance-based feature selection

The maximal information coefficient MIC proposed by [Reshef et al. \(2011\)](#) can measure the linear, nonlinear, functional and non-functional dependencies between variables ([Qin et al., 2023](#)). This demonstrates that MIC can capture a wide range of relationships and is not limited to a specific function type. Among them, functional relationships include linear functional relationships, nonlinear parabolic, periodic, exponential functional relationships, etc., and the superposition of various functions belongs to non-functional relationships. Therefore, the MIC is an effective method to evaluate the relevance between variables ([Liang et al., 2018](#)). Due to the nonlinear nature of Parkinson's voice data ([Zhang et al., 2019a](#)), MIC can be used to measure the relationship between PD voice features and the target value effectively.

Let the PD dataset be represented as $D = \{X, Y\}$, where $X = \{f_1, f_2, \dots, f_i, \dots, f_N\}_{i=1}^N$, $f_i \in \mathbb{R}^{M \times 1}$ represents the feature vector of the sample, N and M represent the feature dimension and the number of samples, respectively. Y denotes the target value, $Y = \{0, 1\}$ for classification and $Y = \{\text{motor-UPDRS}, \text{total-UPDRS}\}$ for regression. Firstly, this paper calculates the MIC score between features and

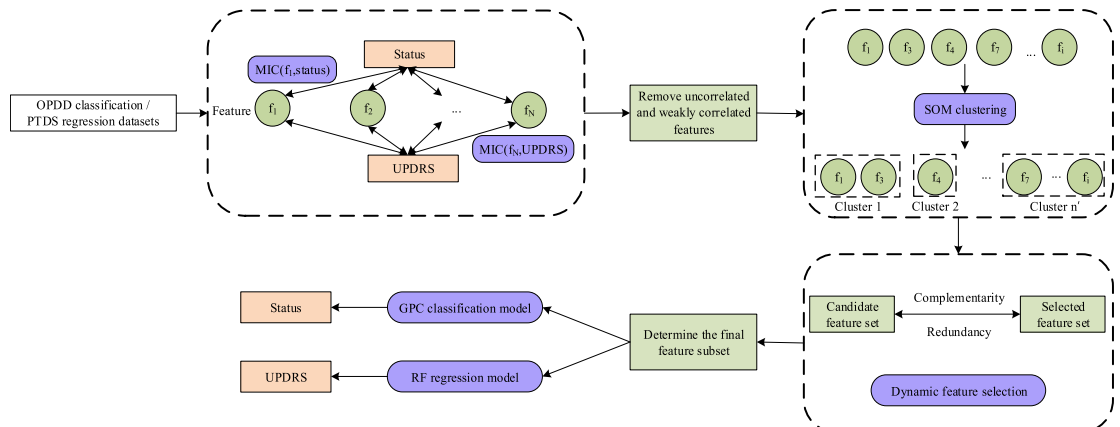


Fig. 1. Schematic diagram of local dynamic feature selection fusion method.

the target value. The main idea of MIC is to divide the grid on the scatter diagram between the feature variable f_i and the target value Y , and calculate the relevance between variable f_i and target value Y in each grid by mutual information. The specific steps are as follows:

Step 1: Grid division. Suppose that all the data between PD voice features and the target value in the dataset is a finite set of ordered pairs $D' = \{(f_k, y_k), k = 1, 2, \dots, M\}$. If the value range of feature f is divided into a segments and the value range of target value y is divided into b segments, the grid of $a \times b$ is defined as G . MIC aims to find all the relationships between variables. The division of variable space will have a great influence on the results of MIC, thus affecting the relevance between features and the target value.

Step 2: Mutual information. After the grid is divided, MIC can calculate the relevance of variables in each grid. The relevance is measured by the mutual information, and the mutual information $MI(f, y)$ between the feature variable f and the target value y in the grid is defined as:

$$MI(f, y) = - \sum_{y_k \in Y} \sum_{f_k \in F} p(f_k, y_k) \log \frac{p(f_k, y_k)}{p(f_k)p(y_k)} \quad (1)$$

Where $p(f_k, y_k)$ represents the joint probability distribution function between f and y , which is approximately obtained from the proportion of the number falling into a certain grid to the sample capacity. $p(f_k)$ and $p(y_k)$ denote marginal probability distribution functions of f and y , respectively.

Step 3: Calculate the maximum MI. By setting different number of divisions a and b in grid G , or changing the location of divisions, and changing the values of a and b at the same time, repeating step1 and step2 will get different mutual information. MIC takes the maximum mutual information value generated under different division methods, denoted as MI' , as the mutual information value of the grid G . The maximum mutual information formula of D' under a given grid G is:

$$MI'(D', a, b) = \max MI(D'|G) \quad (2)$$

Where $D'|G$ represents different distributions of data D' for a given grid G .

Step 4: Normalize the mutual information value. To compare MI' in different grid divisions, it is necessary to normalize them, so as to obtain the feature matrix $C(D')_{a,b}$ of normalized mutual information values under different grid divisions as follows:

$$C(D')_{a,b} = \frac{MI'(D', a, b)}{\log \min\{a, b\}} \quad (3)$$

Step 5: The MIC result. Finally, the maximal information coefficient MIC between the feature variable f and the target value y is defined as:

$$MIC(D') = \max_{a \times b < B(M)} C(D')_{a,b} \quad (4)$$

Where $B(M)$ represents the upper limit value of the division in grid G . The value range of $MIC(D')$ is $[0, 1]$. The larger the MIC value, the stronger the relevance between f and y , that is, f is a strong relevant feature. The smaller the MIC value indicates that the relevance between f and y is weaker, and the f is more likely to be removed. When $MIC=0$, f is irrelevant and should be removed.

2.2. Dynamic feature selection based on redundancy and complementarity

In addition to the relevance between PD voice features and the target value, there may also be strong relevance between features. The high redundancy between features may reduce the efficiency and effectiveness of the model. FS methods often measure the importance of features based on relevance and redundancy, and select the optimal feature subset. However, in some cases, the interaction between highly redundant features and other features may have a positive impact on the model. Therefore, features with high redundancy do not mean that they are meaningless (Li et al., 2019), and it is necessary to consider the complementarity between features.

To select the optimal feature subset, this paper firstly clusters the PD voice features with high relevance by SOM to obtain the candidate features of different clusters. Then, features are dynamically selected from the candidate features according to their relevance, redundancy and complementarity to form the final feature subset. Among them, SOM clustering can divide the candidate features into several clusters, and DFS selects the features of each cluster, which can greatly reduce the number of candidate features

and decrease the complexity of the model.

2.2.1. Self-organizing mapping network clustering

SOM is an unsupervised nonlinear artificial neural network, which can analyze multivariate complex data with multiple dependencies, and is often used for clustering, data analysis and data visualization (Nilashi et al., 2019). To better approximate the complex relevance between PD multivariate voice features (Chen et al., 2021), this paper uses SOM to cluster the voice features with high relevance with the target value, so as to learn and extract the inherent patterns of the feature space. Algorithm 1 shows the process of clustering PD voice features by the SOM.

The SOM network consists of the input layer and the competition layer. Among them, the input layer is the feature vector, and the competition layer is composed of $\text{dim}_x \times \text{dim}_y$ neurons. The SOM uses distance to determine the relevance between feature vectors and neurons, and activates neurons by competitive learning. Only one neuron is activated at each moment, which is called the best matching unit (BMU). Then, the weight vectors of neurons in BMU neighborhood are updated appropriately according to the change of learning rate $\eta(t, d)$. The η decreases continuously as the training time t and the topological distance d between the neurons in the neighborhood and the BMU increase. Therefore, the learning process of SOM updates the weight vectors of the winning neuron and its neighboring neurons according to the similarity measure to make them closer to the input vectors. Thus, the high-dimensional PD voice features can be mapped in the low-dimensional space, and the topological structure of the input data can be discretely represented.

Algorithm 1: PD voice feature clustering based on SOM.

Input: feature vector $X' = \{f_1, f_2, \dots, f_i\}_{i=1}^N$, initial weight vector W of neurons in competition layer, output grid size $\text{dim}_x \times \text{dim}_y$, maximum number of iterations T and learning rate $\eta(t, d)$.
Output: the weight vector W of neurons in the competition layer.
Process:
Initialize W and η , and normalize feature vector X'
for time step $t = 1, \dots, T$ **do**
 Select an input vector $f_i(t)$ from X'
 $dis = 0$
 for neuron $v = 1, \dots, \text{dim}_x \times \text{dim}_y$ **do**
 if $dis = 0$ **then**
 Update the distance between $f_i(t)$ and v :
 $d(f_i(t) - W_v(t)) = \|f_i(t) - W_v(t)\|$
 $dis = d(f_i(t) - W_v(t))$
 BMU: $u = v$
 else
 if $d(f_i(t) - W_v(t)) < dis$ **then**
 $dis = d(f_i(t) - W_v(t))$
 BMU: $u = v$
 Update BMU: $u = v$
 end for
 for neuron $v = 1, \dots, \text{dim}_x \times \text{dim}_y$ **do**
 Update the weight vector: $W_v(t+1) = W_v(t) + \eta(t, d)(f_i(t) - W_v(t))$
 end for
end for

After getting the weight vector W of each neuron in the output layer, the high-dimensional features can be mapped to the two-dimensional plane of the output layer and clustered by determining the shortest distance between the feature vector in X' and each neuron in the output layer. In this method, the similarity of the input PD voice feature data can be expressed as the feature mapping of the nearest neighbors in space by SOM clustering and its topology remains unchanged.

2.2.2. Dynamic feature selection

In traditional FS methods, redundant features are often removed because they may reduce the efficiency and performance of the model. In the subset of features selected using these methods, each feature is status-related and non-redundant with other features. But this is only a necessary but not sufficient condition for generating the optimal feature subset (Chen et al., 2021). This is because the two interdependent features may be complementary and the combination of them may improve the accuracy of the model.

Therefore, in the FS process, this paper not only considers the relevance between features and the target value, but also analyzes the influence of the redundancy among features on the importance of candidate features. In addition, the candidate features with high complementarity with the selected feature set are evaluated by the classifier or regressor. DFS strategy can be used to select the feature that is most complementary to the current feature subset from the candidate features, so that the features with high relevance, low redundancy and high complementarity can be fused into the optimal feature subset. Fig. 2 shows the dynamic feature selection DFS process based on SOM clustering.

As can be seen from Fig. 2, the DFS strategy firstly selects the features with the largest MIC value from each cluster of SOM clustering, sorts these features by the relevance, and adds the features with the largest MIC value to the selected feature subset. Then, in order to select the features with low redundancy and high complementarity from the cluster where the features are located, the proposed DFS strategy combines the relevance $\text{MIC}(f, y)$ between the candidate features f and the target value y , and the average

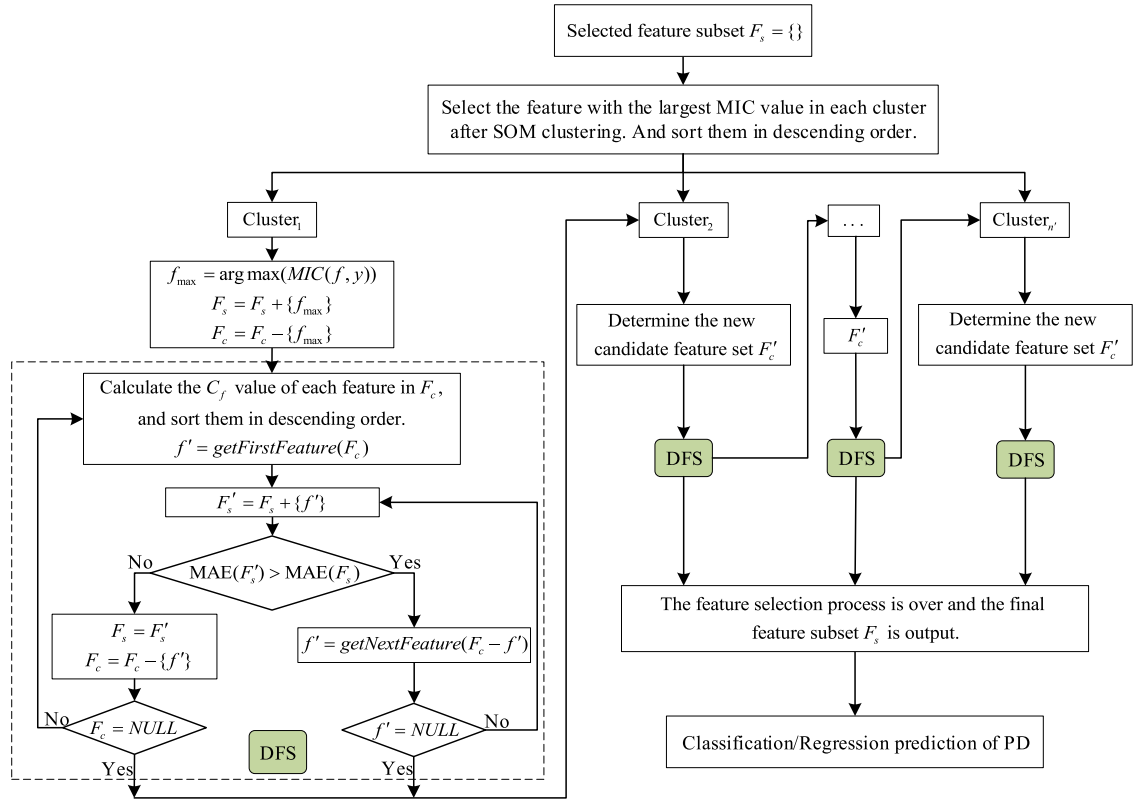


Fig. 2. The dynamic feature selection process based on SOM clustering.

redundancy R_f between the candidate features and the selected subset F_s , which enables to dynamically update the complementarity C_f between the candidate features and the selected subset. Since MIC can also be used as a measure of redundancy between features (Liang et al., 2018), this paper uses MIC to define the relevance between features and the target value, and between features. The redundancy between features f_i and f_j is defined as $MIC(f_i, f_j)$. The larger its value, the smaller the difference of information between features, that is, the stronger the redundancy. The average redundancy R_f and complementarity C_f are defined as follows:

$$R_f = \frac{\sum_{i=1}^{|F_s|} MIC(f, f_i)}{|F_s|} \quad (5)$$

$$C_f = \frac{MIC(f, y)}{R_f} \quad (6)$$

Where F_s represents the selected feature subset, and $|F_s|$ represents the number of features in the set F_s . C_f combines relevance and redundancy to measure the importance of candidate features, and also represents the degree of complementarity between candidate features and the selected subset. The larger C_f , the greater the complementarity between candidate features and the selected subset.

In the current cluster, the dynamic feature selection strategy calculates the complementarity C_f between each candidate feature f and the selected feature set F_s according to Eqs. (5) and (6). Then, the features in the candidate feature set F_c are sorted in descending order according to the C_f value. The feature f with the highest C_f value is added to the selected feature subset F_s to form a new set F_s' . F_s and F_s' are input into the classifier and the regressor, and the mean absolute error (MAE) obtained by them is $MAE(F_s)$ and $MAE(F_s')$ respectively. If $MAE(F_s') \leq MAE(F_s)$, the feature f is added to the selected feature subset F_s , and f is removed from the candidate feature set F_c . If $MAE(F_s') > MAE(F_s)$, the second feature in F_c is added to the set F_s to determine the error. When F_c is empty, all the features in the next cluster are taken as new candidate subsets F'_c , and the features with the greatest complementarity with the selected feature subsets are selected from F'_c and added into the set F_s . The above process is repeated until all the candidate features are tested or the number of features in the selected feature set F_s has reached the upper limit set by the experiment, then F_s is the final output feature subset. Fig. 2 shows the process of dynamic feature selection by testing all candidate features as an example. If the number of selected features reaches the upper limit, the DFS process may end prematurely in any cluster.

2.3. Classification and regression prediction

In this paper, the proposed LDFS framework aims to obtain feature subsets with high relevance, low redundancy and high complementarity by analyzing the relationship between features and the target value, and between features. Among them, in the process of dynamic feature selection, the quality of selected feature subset is evaluated by whether the prediction performance of classifier or regressor is improved. Therefore, the selection of appropriate classifier and regressor has an important impact on the selection of feature subset in the proposed method and on the diagnosis of PD. In PD classification and severity assessment tasks, Gaussian process classification (GPC) and RF regression methods have been proved to be effective respectively (Tsanas et al., 2021; Despotovic et al., 2020).

Gaussian process is a method based on Bayesian and statistical theories, which can effectively use the relevance between data to solve complex classification and regression problems (Jeon and Hwang, 2022). The GPC model predicts the status probability of samples by finding the suitable mapping $F(\cdot)$. And the GPC is a process of finding the appropriate $F(\cdot)$ with the goal of maximizing posterior probability in Bayesian criterion under the assumption that $F(\cdot)$ is a Gaussian process. Distinguishing healthy people from PD patients is a typical classification problem. For this reason, GPC usually converts the predicted output of Gaussian process regression model into status probability through response function to analyze the uncertainty of illness (Despotovic et al., 2020). The GPC can effectively solve complex nonlinear problems, its hyperparameters are easily adjustable, and it has good classification performance, so the GPC is often used in the PD identification task.

RF consists of multiple decision trees and is an integrated learning algorithm (Meghraoui et al., 2021). In the PD severity assessment process, RF randomly selects the PD samples and voice features. Each decision tree selects the best feature to split on and find the optimal tree structure, and then takes the average value of the target value in each subspace as the prediction result of each space. Since the input samples and features of each decision tree are random, the prediction results may be different and RF takes the average of the prediction results of all decision trees as the final prediction value. By randomly sampling PD voice features, RF generates many subspaces containing different voice features. RF uses the results of these subspaces to predict, which increases the diversity of prediction results and thus can improve the prediction performance of the model and effectively reduce the risk of overfitting.

3. Experiment

3.1. Datasets

3.1.1. Classification dataset

The Oxford Parkinson's disease dataset (OPDD) was established by Little et al. (2007). A total of 197 voice samples of 31 subjects were collected. There are 23 PWP and 8 HC. As the measurement of dysphonia is the important feature to distinguish healthy from diseased, 22 linear and nonlinear features are extracted from each piece of voice data, as shown in Table 1.

Among them, the fundamental frequency F_0 defines the frequency of vocal fold vibration per second, and F_{hi} and F_{lo} indicate the maximum and minimum values of the fundamental frequency respectively. Jitter features indicate the degree to which the voice signal deviates from its fundamental frequency. Jitter is mainly caused by uncontrollable vocal fold vibration, which is usually higher in pathological voices. Shimmer features measure the deviation of two consecutive amplitudes, which are often related to breathing and noise, and are caused by reduced vocal fold resistance and vocal fold masses. The noise harmonic ratio (NHR) and harmonic noise ratio (HNR) are often used to measure the quality of a signal. Among them, HNR can represent the noise caused by pathological changes of vocal fold. Compared with HC, the HNR value of PWP tends to be smaller. The human voice system is nonlinear and dynamic. Recurrence period density entropy (RPDE) determines the periodicity of the time series by measuring the repeatability of phase space in the system, thus indicating the ability of the vocal folds to stabilize vocalizations. Because the vocal folds of PWP do not vibrate regularly, their RPDE values are larger. The relevance dimension (D2) measures the irregularity of the reconstructed phase space of the system. The detrended fluctuation analysis (DFA) measures the self-similarity of airflow noise produced by vocal folds. In addition, the nonlinear entropy measures of fundamental frequency variation are also common features of voice signal analysis, including Spread1, Spread2 and pitch period entropy (PPE).

Table 1
OPDD and PTDS datasets.

Dataset	OPDD	PTDS
Number of people	31	42
Number of voice samples	197	5875
Ratio of PWP to HC	23:8	42:0
Gender ratio	Not provided	28 M:14 F
Age distribution	Not provided	36–85
Common features	Jitter (%), Jitter (Abs), Jitter: RAP, Jitter: PPQ5, Jitter: DDP; Shimmer, Shimmer (dB), Shimmer: APQ3, Shimmer: APQ5, Shimmer: APQ11, Shimmer: DDA; NHR, HNR; RPDE, PPE, DFA	
Other features	F_0 , F_{hi} , F_{lo} ; D2, Spread1, Spread2	Age, Sex

3.1.2. Regression dataset

The Parkinson's telemonitoring data set (PTDS) (Tsanas et al., 2009) is from the University of Oxford, which is widely used for PD symptom severity prediction (Ji et al., 2018; Tsanas et al., 2021; Despotovic et al., 2020; Yoon and Li, 2019). The dataset collected the voice data of 42 patients with primary PD, including 28 males and 14 females, and each patient had a history of 5 years. The PD patients need to record 6 persistent vowels /a/ with different loudness on the same day every week, and they can't take medicine during the test. This process lasted for 6 months, and a total of 5875 voice data were collected.

Tsanas et al. (2009) used linear and nonlinear voice signal processing algorithms to extract features from the above voice data, and obtained 16 voice features, as shown in Table 1. As sex and age are also important features in predicting the severity of PD symptoms (Miller and Cronin-Golomb, 2010), they can also be used to track the UPDRS values of patients. Motor-UPDRS and total-UPDRS are often used to evaluate the severity of PD patients. The range of the motor-UPDRS is 0–108, with 0 representing no disability and 108 representing severe motor impairment. The range of total-UPDRS is 0–176, with 0 being healthy and 176 being completely disabled. In this experiment, the maximum values of motor-UPDRS and total-UPDRS are 55 and 41, respectively. Therefore, all the subjects measured are patients with moderate Parkinson's symptoms. Studies have shown that within 5 years, the UPDRS of patients with early PD basically increased linearly without drug treatment (Schüpbach et al., 2010). Therefore, when collecting data, the UPDRS of PD patients to be measured is evaluated at the beginning, after 3 months and after 6 months respectively, and then the UPDRS values corresponding to other times of the patient are obtained on the real UPDRS of these three stages by using piecewise linear interpolation method (Tsanas et al., 2009).

3.2. Experiment setup

During the experiment, the parameters to be adjusted in the proposed LDFS method mainly include the determination of grid division mode in MIC and related parameters in SOM. Among them, the number and way of grid division in MIC method have an important influence on the relevance measurement of features. Therefore, the upper limit $B(M)$ of grid G division in MIC is set as $M^{0.6}$ in this paper, which is proved to be the most effective in practical applications in Reshef et al. (2011). The parameters of SOM are set as follows: learning rate $\eta = 0.9$, output grid size $\dim_x = \dim_y = 4$ and maximum number of iterations $T = 1000$, respectively. To verify the generalization ability of the model, 10-fold cross-validation was used on two datasets, and they were randomly divided into training set, validation set and test set in the ratio of 8:1:1 for experiments. Specifically, the number of samples in the training set, validation set and test set after dividing the OPDD and PTDS datasets are 158, 20, 19 and 4700, 588, 587, respectively. In the process of dynamic feature selection, the training set is used for feature selection, the verification set is used to evaluate the quality of the selected feature subset, and the test set is used to test the performance of the prediction model combining feature selection framework with classifier or regressor. These experiments were run in a processor Intel core i5–12,400 at 2.50 GHz in the windows-10, and were simulated by Python-3.9.

To illustrate the classification performance of the proposed model, accuracy (Acc), sensitivity (Sen) and specificity (Spe) are used to evaluate the results. The formulas are shown in Eqs. (7)–(9). Among them, TP, TN, FP and FN respectively represent the number of true positive, true negative, false positive and false negative. Sensitivity and specificity respectively represent statistical measurement values of positive and negative results for correctly classified Parkinson's disease, respectively.

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (7)$$

$$\text{Sen} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

$$\text{Spe} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (9)$$

To evaluate the performance of the model in predicting the symptom severity of PD and ensure the validity of the prediction results, this paper uses the MAE as the evaluation metric of the prediction model, which is calculated as follows:

$$\text{MAE} = \frac{1}{M} \sum_{k=1}^M |y_k - y'_k| \quad (10)$$

Where M represents the number of test samples, y_k and y'_k represent the true value and the predicted value of the model, respectively. The smaller the MAE, the smaller the deviation between the true and the predicted value, that is, the better the prediction performance.

3.3. Results of the proposed feature selection method

In this paper, the LDFS method selects the optimal subset of features and combines them with the appropriate classifier and regressor to solve the classification and regression problems of PD diagnosis. GPC and RF methods have been proved to be effective on public datasets OPDD and PTDS, so this paper combines LDFS framework with GPC and RF to achieve accurate PD identification and severity prediction, and verifies the feasibility and reliability of the proposed LDFS method. In the 10-fold cross-validation, the dataset divided each time is different, and the LDFS method may obtain different feature subsets due to the influence of data

distribution. To illustrate the proposed method more intuitively, this section shows the experimental results of one of the datasets divided in the cross-validation process.

In the task of PD classification and severity prediction, the LDFS method first analyzes the relevance between features and the target value by the MIC, as shown in Fig. 3. Since the features with low relevance with the target value (status or UPDRS) will reduce the classification and regression performance of the model, this paper sorts the $MIC(f, y)$ of all the features in descending order, deletes the last 20% features with low relevance, and retains the remaining 80% features. Then, the SOM method is used to cluster the retained features. Tables 2 and 3 show the clustering results of SOM on features, respectively. The feature similarity in each cluster is high, and there may be redundant features. The dynamic feature selection process based on SOM clustering combined with GPC and RF can derive the optimal feature subset according to the proposed redundancy and complementarity. Fig. 4 shows the accuracy and MAE variation of the validation set during the dynamic feature selection process. To the best of our knowledge, the simplest features obtained by the existing FS method using the same OPDD and PTDS datasets are 5 and 8, respectively. To compare with the results of the latest literature, the upper limit of the number of features selected by the proposed DFS strategy on OPDD and PTDS datasets is set to 5 and 8, respectively. As shown in Tables 2 and 3 and Fig. 4, when the number of selected feature sets reaches 5 and 8, DFS strategy stops running in cluster 1 and cluster 3, respectively.

3.4. Performance comparison before and after feature selection

Table 4 shows the comparison of PD classification and severity prediction results before and after feature selection on OPDD classification and PTDS regression datasets, and the average value of each evaluation metric is included in the Table 4. Among them, the number of voice features included before and after FS for each sample within the OPDD classification dataset is 22 and 5, respectively. The classifier can be used to distinguish the samples as HC or PWP, and the accuracy, sensitivity and specificity are used as evaluation metrics to evaluate the model. In the classification task, SVM, XGBoost, RF and GPC are tested and compared as classifiers on the same dataset. The results in Table 4 show that the classification performance of GPC method is optimal, which is consistent with Despotovic et al. (2020). Compared with the complete feature set, the LDFS method selected only five features, which improves the accuracy, sensitivity and specificity of GPC algorithm by 5.26%, 4.74% and 2.90% respectively. At the same time, it also improves the accuracy, sensitivity and specificity of other classifiers, which confirms the effectiveness of the proposed LDFS method.

In the PTDS regression dataset, each sample before and after FS contains 18 and 8 features respectively. In this section, the regressors are used to predict the severity of PD, and MAE is used to evaluate the prediction model. Among them, SVR, XGBoost and Gaussian process regression (GPR) are used as baseline methods to compare the prediction results with RF. As can be seen from Table 4, the proposed LDFS method reduces the errors of all regression prediction models as in the classification task. The difference is that the RF model has the lowest error, which indicates that RF is more suitable for analyzing PD severity than GPR.

The number of features selected in the proposed method is small, especially only five features are used in the classification task, which may effectively reduce the computational cost of the method and make the model run faster. Also, it can be seen from Table 4 that the proposed method improves the prediction performance of each classifier and regressor. The performance is also greatly

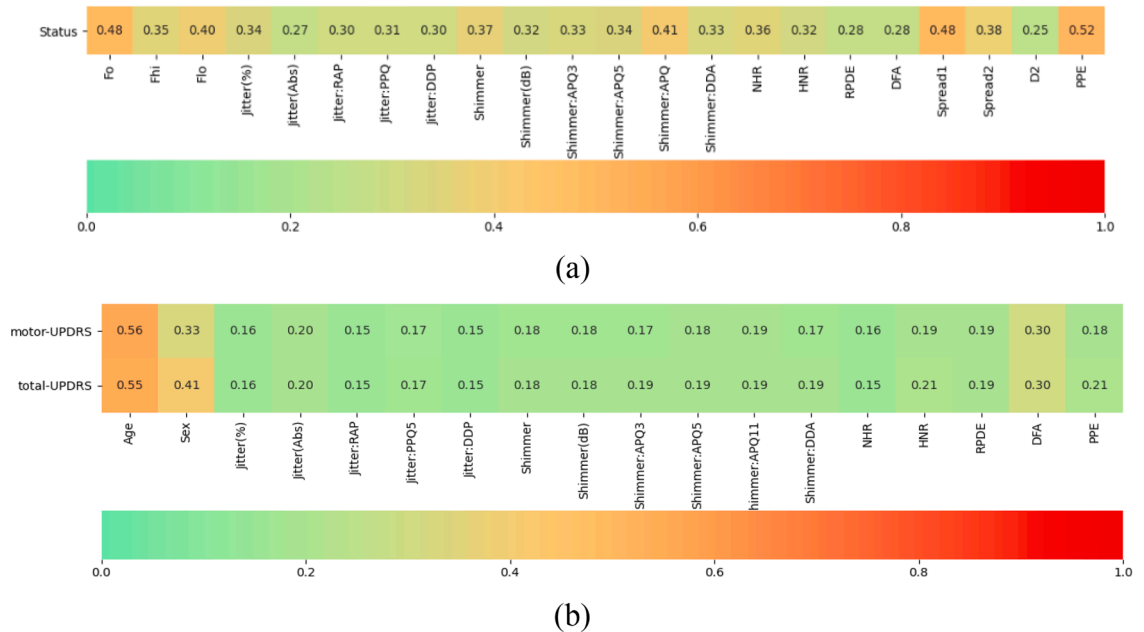


Fig. 3. Relevance (MIC) between features and the target value. (a) Features and Status. (b) Features and UPDRS.

Table 2
Clustering results of SOM in PD classification task.

Cluster	Feature set
1	{Jitter (%), Jitter: RAP, Jitter: PPQ, Jitter: DDP, Shimmer, Shimmer (dB), Shimmer: APQ3, Shimmer: APQ5, Shimmer: APQ11, Shimmer: DDA, NHR, Spread2, PPE}
2	{Spread1}
3	{F ₀ , F _{hi} , F _{lo} }
4	{HNR}

Table 3
Clustering results of SOM in PD regression task.

Cluster	Feature set
1	{Age, HNR, RPDE, DFA}
2	{Sex}
3	{Jitter (Abs), Jitter: PPQ5, Shimmer, Shimmer (dB), Shimmer: APQ3, Shimmer: APQ5, Shimmer: APQ11, Shimmer: DDA}
4	{PPE}

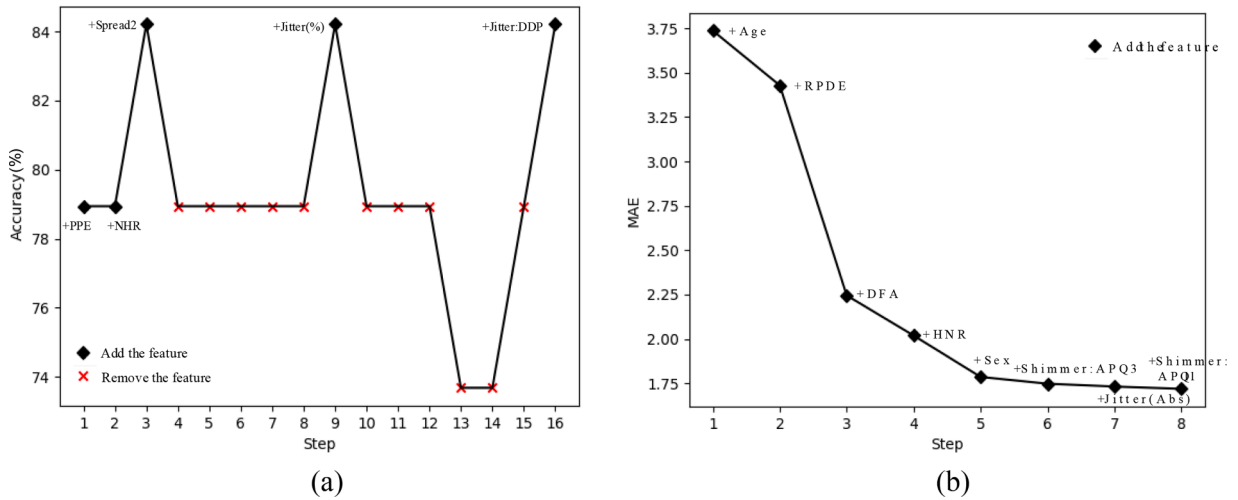


Fig. 4. The dynamic selection process of feature subset. (Each step of the process adds a candidate feature to the feature subset. ◆ indicates that the candidate feature is determined to be added to the subset, and × indicates that the feature added to the subset will be removed). (a) The feature subset (5 features) determined based on GPC is {PPE, NHR, Spread2, Jitter (%), Jitter: DDP}. (b) The feature subset (8 features) determined based on RF is {Age, RPDE, DFA, HNR, Sex, Shimmer: APQ3, Jitter (Abs), Shimmer: APQ11}.

improved for models that perform poorly using the complete feature set, which verifies the effectiveness of the LDFSf method for feature selection based on relevance, redundancy, and complementarity.

3.5. Performance comparison with existing methods

To further verify the effectiveness of the proposed LDFSf framework and its prediction model combined with GPC and RF, this section compares the LDFSf method with the traditional FS method and the results in the recent literature.

Table 5 compares the effects of traditional FS method and LDFSf method on the model performance. Among them, the traditional FS methods include ReliefF, mRMR and Boruta (Tunc et al., 2020). These methods select features based on the relevance between features and the target value and the redundancy between features. The feature selection framework LDFSf in this paper adds complementarity to this foundation to further explore the intrinsic connection between features. To verify the effectiveness of combining relevance, redundancy, and complementarity for feature selection, Table 5 compares the performance impact of different feature selection algorithms on the prediction model with the same classifier and regressor. The GPC and RF in Table 4 show better performance in PD classification and regression prediction, respectively, and the results are consistent with those in Tsanas et al. (2021) and Despotovic et al. (2020). Therefore, in Table 5, each feature selection method is combined with GPC and RF to compare the performance of FS method. In addition, the same number of features is set in this section in order to compare the effectiveness of the selected feature subsets with the same number of features.

As can be seen from Table 5, when the same classifier GPC and regressor RF are used, the LDFSf framework with the combined

Table 4

Comparison of performance results of model classification and prediction of PD severity before and after feature selection.

Method	OPDD dataset			Feature subset			PTDS dataset		Feature subset	
	Total features			(5 features)			Total features		(8 features)	
	Acc (%)	Sen (%)	Spe (%)	Acc (%)	Sen (%)	Spe (%)	Motor-UPDRS	Total-UPDRS	Motor-UPDRS	Total-UPDRS
SVM/SVR	92.31	77.85	97.92	94.98	84.44	99.50	4.71	6.25	3.97	5.30
XGBoost	92.84	83.22	97.19	95.99	88.89	99.50	1.88	2.32	1.73	2.15
RF	89.23	70.82	95.51	95.87	74.44	98.23	2.93	3.75	1.67	2.06
GPC/GPR	93.29	87.83	96.70	98.20	92.00	99.50	1.75	2.17	1.62	1.99

model of both have obtained the optimal performance in PD identification and severity progression prediction, respectively, which indicates that it is necessary to combine relevance, redundancy and complementarity to analyze the intrinsic relationship between PD voice features.

In addition, Table 6 compares the results of the proposed LDFS method with those of the existing literature. From the perspective of using the same dataset with different research methods, the results reported in Senturk (2020), Liu et al. (2021), Despotovic et al. (2020), Peker and Kubat (2021) and in Ji et al. (2018), Tsanas et al. (2021), Despotovic et al. (2020), Yoon and Li (2019) are selected for comparison with the results of the proposed LDFS method on the OPDD and PTDS datasets, respectively. Among them, the FS method (GPC-ARD and GPR-ARD) based on automatic relevance determination proposed in Despotovic et al. (2020) is combined with Gaussian process for PD identification and severity prediction, respectively. As shown in Table 6, among all the compared literature, the method proposed in Despotovic et al. (2020) has the highest accuracy and the lowest MAE error in PD classification and severity progress prediction, respectively. However, the prediction performance of the proposed LDFS method has better results than Despotovic et al. (2020) on both datasets, which shows the superiority of the proposed method.

4. Discussion

In this study, an effective feature selection method LDFS is proposed to achieve PD classification and severity progression prediction by combining classifier GPC and regressor RF. Experiments on two datasets verify the effectiveness of the proposed method, thus demonstrating its importance for PD identification and early diagnosis and treatment of patients.

4.1. Effectiveness of MIC

The LDFS method selects the final feature subset based on relevance, redundancy and complementarity. Among them, redundancy and complementarity can be expressed by the relevance between features, and between features and selected feature subsets, respectively. Therefore, the selection of an appropriate relevance evaluation method has an important impact on the measure of feature importance and the result of feature subset. Existing methods commonly use the Spearman correlation coefficient to measure relevance (Tsanas et al., 2021; Tsanas et al., 2009), but it is more suitable for linear or monotonic relationships, and the reliability of the results will be greatly reduced in the face of more complex relevance relationships. However, the MIC can capture a wide range of relationships among variables, including linear, nonlinear, functional and non-functional relationships, etc. In addition, the MIC can get similar scores for different types of noise data, which has good robustness (Reshef et al., 2011).

As shown in Fig. 3(a), the relevance between PPE and status is the highest when the relevance between features and status is tested, which is consistent with the conclusion of Little et al. (2009). In Fig. 3(b), it can be observed that there is a strong relevance between age and sex and the target value by calculating the MIC value between the features and UPDRS. In addition, features related to vocal fold vibration such as DFA and RPDE also have high relevance with UPDRS (Tunc et al., 2020). Therefore, MIC can effectively identify the features with the high relevance with the target value, and these features are included in the final feature subset, thus laying a foundation for improving the performance of PD classification and severity progression prediction.

Table 5

Comparison of the results between the proposed method and the traditional feature selection methods.

FS method	GPC classifier			RF regressor	
	Acc(%)	Sen(%)	Spe(%)	Motor-UPDRS	Total-UPDRS
ReliefF	92.23	90.60	98.76	1.80	2.12
mRMR	92.31	90.63	98.78	1.78	2.12
Boruta	89.74	89.38	91.43	1.82	2.19
LDFS(ours)	98.20	92.00	99.50	1.62	1.99

Table 6

Performance comparison between the proposed method and the existing methods.

Method	OPDD dataset			Method	PTDS dataset	
	Acc (%)	Sen (%)	Spe (%)		Motor-UPDRS	Total-UPDRS
Senturk (2020)	93.84	–	–	Ji et al. (2018)	3.00	4.60
Peker and Kubat (2021)	88.03	91.84	–	Yoon and Li (2019)	2.40	–
Liu et al. (2021)	89.17	–	–	Tsanas et al. (2021)	2.65	3.17
Despotovic et al. (2020)	96.92	90.00	99.29	Despotovic et al. (2020)	1.65	2.02
LDFSf (ours)	98.20	92.00	99.50	LDFSf (ours)	1.62	1.99

4.2. Effect of SOM clustering

To illustrate the impact of SOM clustering on the prediction model, [Table 7](#) shows the performance of the proposed LDFSf model before and after using SOM clustering and the results of time consumption.

As can be seen from [Table 7](#), the running time of the prediction model is reduced on both datasets after SOM clustering. This is due to the fact that after SOM clustering, the similarity of features in the same cluster is high, and the similarity of features between clusters is low. In the process of dynamic feature selection, the proposed method only pays attention to the relationship between candidate features in the same cluster and feature subsets, and avoids considering a large number of irrelevant features in other clusters, thus reducing the computational cost of the model. In addition, the running time of the LDFSf method before and after SOM clustering on the OPDD dataset is smaller than that on the PTDS dataset because the sample size of the OPDD is smaller than that of the PTDS dataset.

For the prediction results, the performance of the model after SOM clustering is improved in both classification and regression prediction, which is consistent with the results of [Nilashi et al. \(2018\)](#), [Nilashi et al. \(2019\)](#) and [Nilashi et al. \(2020\)](#). The reason for this result may be that SOM clustering can find the relevance between features, and the dynamic feature selection methods combining redundancy and complementarity on this basis can better analyze the intrinsic connections between features. Specifically, SOM clusters based on the similarity between features, and the high similarity of features in each cluster implies that there may be a large redundancy between features. For example, cluster 3 in [Table 3](#) contains features that are all linear and have high relevance between features ([Despotovic et al., 2020](#)). The way they measure voice signals is very similar, so some redundant features may need to be removed. However, the dynamic feature selection method based on SOM clustering can not only reduce the calculation cost, but also help to remove redundant features and improve the prediction performance of PD classification and regression models.

4.3. Analysis of results

As can be seen from [Table 5](#), compared with the traditional FS algorithm, the proposed LDFSf method shows the optimal prediction performance in PD classification and severity progression prediction tasks. Among them, the ReliefF method only considers the feature relevance without analyzing redundancy. The mRMR method is used to select a set of features with high relevance with the target value and low redundancy among variables. The Boruta method aims to find out all the feature with relevance to the target value. ReliefF, mRMR and Boruta determine the importance of features based on the relevance between features and the target value and the redundancy between features, but do not consider the complementarity between candidate features and feature subsets. The proposed LDFSf framework combines relevance, redundancy and complementarity for feature selection, so the proposed prediction model shows superiority with the same classifier GPC and regressor RF in [Table 5](#).

Moreover, as shown in [Table 6](#), the prediction results of the LDFSf model have better performance than those of the compared literature. For classification, [Peker and Kubat \(2021\)](#) put forward an ensemble classification algorithm that discretizes continuous variables without feature selection, in which irrelevant or redundant features may increase the prediction error of the classification algorithm. In order to improve the prediction performance of the model, [Senturk \(2020\)](#) and [Liu et al. \(2021\)](#) proposed different feature selection methods based on the correlation of features and the elimination of high noise and high aliasing in PD speech, respectively. For regression, [Ji et al. \(2018\)](#) and [Yoon and Li \(2019\)](#) used transfer learning to make up for the small data of PD patients by evaluating the similarity between patients. However, it is not easy to measure the similarity of PD patients in different time dimensions, which reduces the prediction performance of the model. Considering the class imbalance problem widely existing in the data ([Li et al., 2023](#)), [Tsanas et al. \(2021\)](#) utilized four FS methods and combined them with RF for PD severity assessment. This method may be affected by bandwidth restriction or channel transmission errors, resulting in the loss of accuracy in UPDRS. Among all the compared literature, the classification and regression model of Gaussian process based on ARD ([Despotovic et al., 2020](#)) achieves better prediction results while reducing the number of features. However, ARD method calculates the weights of individual features and selects the feature with higher weight to form a feature subset without considering redundancy and complementarity. The results in [Table 6](#) show that the experimental results of the LDFSf method outperform the proposed by [Despotovic et al. \(2020\)](#), thus verifying the effectiveness of feature selection based on relevance, redundancy and complementarity. According to the above results, the LDFSf method can select a small and effective subset of PD voice features, which has certain reference value to help researchers more accurately perform PD classification and severity assessment.

Table 7

Comparison of effects of SOM clustering on prediction model performance and time consumption.

Dataset	Evaluation metrics		Not used SOM	Use SOM
OPDD	Acc (%)		94.74	98.20
	Sen (%)		91.23	92.00
	Spe (%)		99.38	99.50
	Time (s)		10.84	7.56
PTDS	Motor-UPDRS	MAE	1.64	1.62
		Time (s)	259.15	258.77
	Total-UPDRS	MAE	2.14	1.99
		Time (s)	259.07	258.77

4.4. Limitations

Although the LDFS method can select the effective feature subset and improve the prediction accuracy, it still has some limitations. First, the rough extraction of traditional voice features from the whole voice data makes the model accuracy limited. The more representative features of PD voice signal, such as time-frequency mixing (Zhang et al., 2021b; Karan et al., 2021) and energy direction features (Zhang et al., 2023), need to be considered, which will be analyzed experimentally in the future research. Second, the number of experimental samples is limited. Although the performance of the method can be reflected to some extent, but more data will be more convincing. In recent years, causal feature selection has gradually attracted greater attentions for its ability to discover the underlying mechanism of a system (Yu et al., 2020). Specifically, the causal feature selection method can select the causal features by learning the Markov boundary of the target variable, which can improve the robustness of the model (Guo et al., 2022). Therefore, future work will continue to expand the dataset, extract more effective features and introduce causal feature selection methods to further improve the prediction performance and robustness of the model.

5. Conclusion and future work

This paper has proposed a local dynamic feature selection fusion method LDFS for PD classification and symptom severity prediction by analyzing the relevance, redundancy and complementarity among PD voice features. The proposed LDFS method uses the maximal information coefficient to determine the relevance between features and the target value, and selects feature subsets with high relevance, low redundancy and high complementarity by the dynamic feature selection method based on SOM clustering. The experimental results show that the proposed LDFS method can select a small and effective feature set in the real public OPDD and PTDS datasets, so that the optimal prediction performance is obtained with the appropriate GPC classifier and RF regressor.

In the future research, more raw voice samples will be collected, and more distinguishing features extracted by scholars will be tried to expand the feature set. This may enhance the prediction performance of the model and facilitate the possibility of exploring other neural network-based clustering algorithms. Moreover, the causal feature selection method will be introduced to learn Markov boundary of the target variable UPDRS, so as to select voice features that have causal relationship with UPDRS to improve the robustness and practicability of the diagnosis model. Further, the cross-database study will also be an interesting and challenging direction.

CRediT authorship contribution statement

Zaifa Xue: Investigation, Methodology, Writing – original draft, Validation. **Huibin Lu:** Project administration, Supervision, Writing – review & editing. **Tao Zhang:** Conceptualization, Methodology, Funding acquisition, Project administration, Supervision. **Jiahui Xu:** Methodology, Validation, Writing – review & editing. **Xiaonan Guo:** Investigation, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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